EVOLUTION OF EMBODIED AFFECT

Complete catalog map · Human foundations, robot expression and embodied interaction

80 W entries + 14 H foundations / method-background entries + 3 R resources = 97 unique placements

Five-stage loop / six technical modules • Orange rows: first-stage research focus • Each ID appears exactly once

Time = verified first public date when available; otherwise dated publication/source. Earlier decades are compressed. No performance or maturity axis.

≤2022 · 14 entries — **2023 · 4 entries** — **2024 · 12 entries** — **2025 · 35 entries** — **2026 · 32 entries**

01 Perceive human affect

Situated face, voice, body and touch cues

- **W072** 2024 — **SeoulTech FER-to-motion** — SeoulTech HCIR — Korean Robotics Society
- **W027** 2024-10-24 — **UGotMe** — Macquarie × SYSU × SJTU — pub 2025 · ICRA
- **W056** 2026-07-20 — **EmArm** — HKUST × Shenzhen × Nankai — Nature Sensors

02 Understand & decide

Context, response target and timing

- **W019** 2024 — **Nadine** — Geneva MIRALab — preprint
- **W053** 2024 — **MHE** — SJTU RL² — TAFFC
- **W015** 2025 — **Nikola instruction styles** — RIKEN × Kyoto × NII — Frontiers RAI
- **W032** 2025 — **HIAER** — UCL — preprint
- **W007** 2026 — **Context-Aware** — Stuttgart SIR — HRI
- **W038** 2026 — **SeM²** — Wuhan University — preprint
- **W040** 2026 — **Holistic Haru** — Honda Japan — HRI Companion
- **W073** 2026-03-19 — **RG-VLMD** — Liverpool · affiliations provisional — preprint

3A Affect-conditioned generation

Task content + affect / social expression

- **W003** 2023-12-11 — **Alter3** — Tokyo × Alternative Machine — pub 2025 · Frontiers RAI
- **W002** 2024-10-30 — **EMOTION** — Apple — pub 2025 · RA-L
- **W031** 2025-05-16 — **SLM text + gesture** — Uppsala — Frontiers RAI
- **W041** 2025-08-12 — **Honda emotion flow** — Honda Europe — RO-MAN
- **W005** 2025-12-02 — **PEPE** — KITECH × UST — Applied Sciences
- **W078** 2026 — **Emotion Retargeting** — Honda Europe — ICRA workshop
- **W009** 2026-08-28 — **PAMoR** — UCL — preprint · RA-L under review
- **W008** 2026-09-20 — **EmoPose** — HKUST(GZ) × Shandong × RoboScience — preprint
- **W010** 2026-09-27 — **SocialHumanoid** — SJTU × ZTE — preprint

3B Physical realization

Tracking, task constraints, expression retention

- **W046** 2025-02-16 — **DFM** — ETH × Sony — ICRA
- **W036** 2025-05-26 — **AMOR** — Disney × UFRGS — SIGGRAPH
- **W035** 2025-10-18 — **Operator Imitation** — Disney Research — IROS
- **W069** 2026 — **EmoLo** — Osaka × Kobe — RO-MAN
- **W024** 2026-01-24 — **ReSIn-HR** — XJTLU — preprint
- **W037** 2026-05-13 — **Olaf** — Disney Research — RA-L
- **W076** 2026-10-14 — **Beyond Kinematic Tracking** — UCL — preprint
- **W034** 2026-10-19 — **RoboMMT** — Stuttgart SIR — preprint

04 Human interpretation & feedback

Recognition, appropriateness and interaction outcomes

- **W020** 2023-12-15 — **VITA** — Cambridge × Politecnico di Milano — pub 2025 · THRI
- **W021** 2025 — **Educational Empathy** — XJTLU × Liverpool — Frontiers RAI
- **W028** 2025 — **Robot-Led Reappraisal** — Cambridge — preprint
- **W029** 2025 — **Embodiment × Empathy** — VU Amsterdam — ICSR
- **W022** 2026 — **AffectLoop** — Kyoto × Boise State — APSIPA · preprint
- **W023** 2026 — **Fluid-Xpress** — BLCU × Manchester × SHU × Penn State — HRI Companion
- **W025** 2026 — **MiRo-E** — Nottingham CHART — Sensors
- **W030** 2025 — **Beyond the monotonic** — Humboldt Berlin — CHB: AH
- **W074** 2025 — **BiRDe support** — BGU / TU Wien collaboration — CHB: AH
- **W075** 2025 — **Emotion Appropriateness** — BGU / TU Wien collaboration — HRI
- **W064** 2025-10-06 — **Robot empathy intervention** — CUHK collaboration — JADD

05 Refine & adapt

Correction, preferences and bounded adaptation

- **W001** 2024-01-26 — **GenEM** — DeepMind × Hoku Labs — HRI
- **W013** 2025-03-21 — **HAPI** — RIKEN × Kyoto × NII — IROS
- **W042** 2026-02-19 — **Proxy-based optimization** — Monash collaboration — RA-L

FB Robot body foundations

Motion parameters, design, latent spaces and control

- **H005** 2000-05 — **Kismet** — MIT · Breazeal — historical source
- **H006** 2015-02-04 — **Shimi** — Georgia Tech × IDC · Bretan / Hoffman / Weinberg — historical source
- **H007** 2019-03-27 — **Task-Constrained Laban** — UIUC × dance collaborators — historical source
- **H009** 2020 — **Target VA-CVAE** — SFU ROSIE collaboration — historical source
- **H008** 2020-03 — **MoveAE** — Cornell × Honda Japan — historical source
- **H010** 2022-03-04 — **Functional Expressive Motion** — UC Berkeley — historical source
- **W012** 2024 — **Face2Gesture** — Cornell / CMU / Plus One — THRI
- **W066** 2024-01 — **Dynamic face / body model** — UNIST — JCROS
- **W006** 2025 — **ELEGNT** — Apple — preprint
- **W016** 2025 — **Touched by ChatGPT** — Ghent × imec — HRI
- **W017** 2025 — **Culture-specific touch** — Ghent × imec — preprint
- **W018** 2026 — **BioAI²** — Southeast University × NUS — Nature Communications
- **W045** 2026 — **Swarm storytelling** — Tsinghua MiLab × Neurobo — IUI
- **W050** 2026 — **RoboBunting** — UBC SPIN — HRI
- **W039** 2025 — **Laban Shape** — CU Boulder collaboration — RO-MAN
- **W055** 2025 — **Gait-to-Gait** — XJTLU × UniSC — Sensors
- **W067** 2025 — **Adaptive Emotional Expression** — UNIST — ICRA
- **W068** 2026 — **Dynamic Modulation** — UNIST — ICRA
- **W071** 2026 — **Arm motion qualities** — Oregon State SHARE — RA-L
- **W077** 2026-03-10 — **Dance2Hesitate** — CU Boulder × PLU — HRI workshop · dataset / workshop
- **W079** 2025 — **MOFU** — Sony + collaborators — RSJ
- **W033** 2025-04-09 — **Expressive DMPs** — Stuttgart SIR — IROS
- **W004** 2025-05 — **GPT-Driven Gestures** — Monash collaboration — RA-L
- **W052** 2026-03-12 — **MojiKit** — HKUST(GZ) / HKUST × CAS + partners — CHI
- **W070** 2026-07-15 — **Anatomy of Uncertainty** — NTU · affiliations provisional — preprint

FF Robot face & head foundations

Actuation, expression mapping and timing

- **W011** 2024 — **Facial coexpression** — Columbia collaboration — Science Robotics
- **W060** 2024 — **AU-FEDS** — Hohai × Changzhou × Leicester × Manchester — T-RO
- **W014** 2025 — **FABG** — USST × Droid × Hamburg — IROS
- **W057** 2025 — **Viseme consistency** — Harbin Institute of Technology — ICCRE
- **W061** 2025 — **PVC-gel face** — ZJUT · bibliography pending — ICRA · report · pending
- **W049** 2026 — **Expressive Furhat** — Ghent × imec — HRI Companion
- **W043** 2026-02-02 — **Facial emotion enabling** — USST × Mech-Mind × CETC32 × Hamburg — Concurrency Comput.
- **W058** 2026-03-07 — **Pengrui** — BIT collaboration · affiliations provisional — preprint
- **W051** 2025-04-01 — **Implicit-emotion talking head** — USST — Applied Computer Research
- **W044** 2025-07-22 — **Morpheus** — Tsinghua × BAAI × MGI × Beihang — RSS
- **W059** 2025-10-27 — **YU-F01** — HUST collaboration — ICIRA
- **W065** 2026-03-13 — **Face calibration** — CAS-related Taicang computing institute — patent CN121661700A
- **W062** 2026-07-10 — **Facial expression survey** — Shanghai University — Computer Science · survey

FH Human body & theory

Perception, psychology and performing arts

- **H001** 1973-06 — **Biological motion** — Uppsala · Johansson — historical source
- **H002** 1980 — **Affect circumplex** — UBC · Russell — historical source
- **H003** 1989-12 — **Body motion & emotion** — Tilburg · de Meijer — historical source
- **H004** 2004-11-16 — **Whole-body affect perception** — MGH / Harvard × Tilburg · de Gelder et al. — historical source
- **W054** 2024-11-08 — **Body expression & eye tracking** — Nanjing Forestry University — Biomimetics
- **W047** 2024-11-26 — **Robots and Dance** — RAD Lab · Amy LaViers — pub 2025 · Annual Review
- **W048** 2025 — **The Body in Affective Robotics** — Canberra × RAD × Penn State / Stelarc — pub 2026 · TAFFC · review
- **W063** 2025 — **Motion arousal perception** — Guangdong University of Technology — Applied Sciences

FD Virtual humans & motion data

Transferable data / representations; physical gap remains

- **R001** 2022-03-10 — **BEAT** — Tokyo × Keio × Huawei × JAIST — ECCV
- **R002** 2023-12-31 — **EMAGE / BEAT2** — Tokyo / Keio / MPI / JAIST / Tsinghua — pub 2024 · CVPR
- **R003** 2025-10-11 — **EmoDiffGes** — East China Normal University — CGF / PG

EL Evidence-limited entries

Abstract / engineering / white-paper evidence

- **W080** 2025 — **Intempus** — Intempus — white paper
- **W026** 2026 — **AIM Multimodal** — Author affiliations pending — AIM abstract · abstract · pending

AI AI / LLM background

Context, not a necessary dependency of every method

- **H014** 1997-09-24 — **Affective Computing** — MIT · Picard — method / discipline background
- **H011** 2017-06-12 — **Transformer** — Google / Toronto author collaboration — method / discipline background
- **H013** 2023-03-15 — **GPT-4 Technical Report** — OpenAI — method / discipline background
- **H012** 2020-06-19 — **DDPM** — UC Berkeley — method / discipline background

READING THE MAP

Rows are editorial primary placements, not mutually exclusive capabilities. W041 / W078 are related works, not two automatically independent breakthroughs.

AI context: ChatGPT research preview, 2022-11-30 (shown here separately; not counted among the 97 indexed entries).

Record types remain distinct: research / review / dataset / workshop / patent / white paper / limited evidence. Full metadata and scholars: README.

Coverage audited: W001–W080 + H001–H014 + R001–R003. Click a card in the SVG / HTML to open its original source.